MODULE 1

Q No	
1	**Explain the Role of Mechanical Engineering in Industries and Society**
	• **Automotive Industry**: Mechanical engineers contribute to the design and development of automobiles, including the engine, chassis, suspension, and other critical components. They also play a role in manufacturing processes and quality control. • **Aerospace Industry:** Mechanical engineers are involved in designing and manufacturing aircraft and spacecraft. • **Energy Industry**: Mechanical engineers contribute to design and optimize power plants, turbines, engines, and renewable energy systems. They work on energy storage, efficiency improvement, and environmental sustainability. • **Manufacturing Industry**: Mechanical engineers are involved in the design and optimization of manufacturing processes and equipment. They work on improving productivity, quality, and cost-effectiveness in production lines. • **Robotics and Automation**: Mechanical engineers contribute to the development of robotic systems and automation technologies. They design robotic mechanisms, grippers, and control systems. • **Consumer Products**: Mechanical engineers are involved in the design and development of various consumer products, including appliances, electronic devices, sporting equipment, and more. **Role of Mechanical Engineers in Society** • **Transportation**: Mechanical engineering is largely concerned with road transportation, and networks, rail and subway systems, airports, and shipping ports. • **Medical**: Engineer's responsibilities in medicine can include research, development, testing and evaluation of medical devices, advisement on new biomedical purchases for hospitals and medical centers. • **Agriculture**: The graduates are trained professionals who understand technology and can solve problems in the areas of agricultural production and management of rural areas with a special focus on machinery. • **Defense:** Manage the teams of skilled technicians that monitor, maintain and repair formidable military hardware such as tanks, artillery guns and armored logistic vehicles. Not all types of entry are open for this job. • **Power generation**: Mechanical engineers also work with the operations of the plant.
2	**Explain the Emerging Trends and Technologies in Automotive and Aerospace sectors**

	• **AUTOMOTIVE SECTOR** 1. **Electric Vehicles (EVs):** The automotive industry is experiencing a shift towards electric vehicles powered by batteries or hydrogen fuel cells, reducing dependence on fossil fuels and minimizing emissions. 2. **Autonomous Vehicles**: Self-driving or autonomous vehicles are being developed with advanced sensors, machine learning algorithms, and connectivity technologies. 3. **Connected Cars**: Connectivity and communication technologies are being integrated into vehicles, enabling features such as real-time traffic information, remote diagnostics, and vehicle-to-vehicle communication. • **AEROSPACE SECTOR** 1. **Lightweight Materials**: The aerospace industry is exploring advanced lightweight materials, such as carbon fiber composites and titanium alloys, to reduce weight and improve fuel efficiency in aircraft. 2. **Electric Propulsion**: Electric propulsion systems, including electric motors and batteries, are being developed for aircraft. 3. **Advanced Aerodynamics**: Innovations in aerodynamics, including improved wing designs, laminar flow control, and adaptive structures, aim to enhance aircraft performance and fuel efficiency • **MARINE SECTOR** 1. **Alternative Fuels**: The marine industry is exploring alternative fuels like liquefied natural gas (LNG), biofuels, and hydrogen to reduce emissions and comply with environmental regulations. 2. **Autonomous Vessels**: Autonomous or unmanned vessels are being developed for various applications, including cargo shipping, surveying, and offshore operations. 3. **Sustainable Ship Design**: The focus on sustainability in the marine sector includes the development of eco-friendly ship designs, such as energy-efficient hull shapes, hybrid propulsion systems, and improved waste management systems.
3	Write a short note on i) Global warming ii) Ozone layer depletion
	Global warming The gradual heating of Earth's surface, oceans and atmosphere, is caused by human activity, primarily the burning of fossil fuels that pump carbon dioxide (CO_2), methane and other greenhouse gases into the atmosphere. Global warming is the long-term rise in the average temperature of the Earth's climate system **Causes for Global Warming:** Global warming occurs when carbon dioxide (CO_2) and other air pollutants and greenhouse gases collect in the atmosphere and absorb sunlight and solar radiation that have bounced off the earth's surface. Normally, this radiation would escape into space—but these pollutants, which can last for years to centuries in the atmosphere, trap the heat and cause the planet to get hotter. That's what's known as the greenhouse effect.

	Effects of Global warming Ocean Warming, Ice Loss, change in climate, Sea level rise, Depletion of ozone layer, Causes for unseasonal rains, Causes for different types of skin diseases **Ozone Layer Depletion:** The ozone layer is an invisible layer of protection around the planet that protects us from the Sun's harmful rays (Ultra-Violet Rays). In other words, it acts as nature's umbrella against UV radiations. Ozone depletion is the thinning of earth's ozone layer. This is caused due to the presence of greenhouse gases in the atmosphere. Depletion of the ozone layer is mainly due to the interaction of ozone with chlorine and bromide present in Chlorofluorocarbons (CFC's). Once these toxic gases reach the upper atmosphere, they cause a hole in the ozone layer causing its depletion. This is one of the most important current environmental challenges. **Effects of Ozone Layer Depletion:** • Increases the flux of UV radiations over earth's biosphere • Causes skin cancer and eyes disorders • Decreases the rate of photosynthesis • Leads to greenhouse effect • It causes decrease in the yield of crops • Adversely affects human immunity system and causes genetic abnormalities • Causes the degradation of plaints, plastics and other polymer materials • It affects the marine life
4	Explain briefly the emerging trends of Mechanical engineering in Manufacturing & Energy sectors. • **ENERGY SECTOR** 1. **Renewable Energy**: There is a growing focus on renewable energy sources such as solar, wind, hydro, and geothermal power. 2. **Energy Storage:** The development of advanced energy storage technologies, such as lithium-ion batteries, flow batteries, and hydrogen storage, is crucial for managing the intermittent nature of renewable energy sources and enabling grid flexibility. 3. **Smart Grids**: Smart grid technologies incorporate digital communication and control systems into traditional electrical grids. They enable more efficient electricity distribution, demand response mechanisms, and integration of renewable energy sources into the grid. • **MANUFACTURING SECTOR** 1. **Additive Manufacturing (3D Printing):** Additive manufacturing allows the production of complex parts and prototypes using layer-by-layer material deposition. It offers advantages such as design flexibility, reduced material waste, and rapid prototyping, transforming manufacturing processes.

	2. **Industrial Internet of Things (IIoT):** IIoT involves the integration of sensors, devices, and data analytics to optimize manufacturing processes, monitor equipment performance, enable predictive maintenance, and enhance overall productivity. 3. **Artificial Intelligence (AI) and Machine Learning**: AI and machine learning are being used to improve automation, optimize production lines, and analyze vast amounts of data for predictive analytics and quality control in manufacturing.
5	Outline the following i) Environmental Issues ii) Bio fuels **i. Environmental Issues** Environmental issues are harmful effects of human activity on the biophysical environment. The various environmental issues are global warming, ozone layer depletion, air pollution, water pollution, natural resource depletion, waste disposal. Climatic changes, deforestation, acid rain etc. **Green House Effect**: The greenhouse effect is a natural process that warms the Earth's surface. When the Sun's energy reaches the Earth's atmosphere, some of it is reflected to space and the rest is absorbed and re-radiated by greenhouse gases. Greenhouse gases include water vapor, carbon dioxide, methane, nitrous oxide, ozone and some artificial chemicals such as chlorofluorocarbons (CFCs). The absorbed energy warms the atmosphere and the surface of the Earth. Step 1: Solar radiation reaches the Earth's atmosphere - some of this is reflected into space. Step 2: The rest of the sun's energy is absorbed by the land and the oceans, heating the Earth. Step 3: Heat radiates from Earth towards space. Step 4: Some of this heat is trapped by greenhouse gases in the atmosphere, keeping the Earth warm enough to sustain life. Step 5: Human activities such as burning fossil fuels, agriculture and land clearing are increasing the amount of greenhouse gases released into the atmosphere. Step 6: This is trapping extra heat and causing the Earth's temperature to rise. **Global warming** The gradual heating of Earth's surface, oceans and atmosphere, is caused by human activity, primarily the burning of fossil fuels that pump carbon dioxide (CO_2), methane and other greenhouse gases into the atmosphere. Global warming is the long-term rise in the average temperature of the Earth's climate system **Causes for Global Warming:** Global warming occurs when carbon dioxide (CO_2) and other air pollutants and greenhouse gases collect in the atmosphere and absorb sunlight and solar radiation that

	have bounced off the earth's surface. Normally, this radiation would escape into space—but these pollutants, which can last for years to centuries in the atmosphere, trap the heat and cause the planet to get hotter. That's what's known as the greenhouse effect. **Effects of Global warming** Ocean Warming, Ice Loss, change in climate, Sea level rise, Depletion of ozone layer, Causes for unseasonal rains, Causes for different types of skin diseases **Ozone Layer Depletion:** The ozone layer is an invisible layer of protection around the planet that protects us from the Sun's harmful rays (Ultra-Violet Rays). In other words, it acts as nature's umbrella against UV radiations. Ozone depletion is the thinning of earth's ozone layer. This is caused due to the presence of greenhouse gases in the atmosphere. Depletion of the ozone layer is mainly due to the interaction of ozone with chlorine and bromide present in Chlorofluorocarbons (CFC's). Once these toxic gases reach the upper atmosphere, they cause a hole in the ozone layer causing its depletion. This is one of the most important current environmental challenges. **Effects of Ozone Layer Depletion:** • Increases the flux of UV radiations over earth's biosphere • Causes skin cancer and eyes disorders • Decreases the rate of photosynthesis • Leads to greenhouse effect • It causes decrease in the yield of crops • Adversely affects human immunity system and causes genetic abnormalities • Causes the degradation of plaints, plastics and other polymer materials • It affects the marine life

ii) Bio fuels

Biofuels

Biofuels are energy sources made from living things, or the waste that living things produce. It is the renewable energy source. Biofuels are combustible fuels created from biomass. There are two main types of biofuels one is **ethanol** and another **biodiesel**.

Ethanol

Ethanol can be produced by the action of microorganisms and enzymes through the fermentation of sugar cane. Ethanol can be used in petrol engines as a replacement for gasoline; it can be mixed with gasoline to any percentage. Most existing car petrol engines can run on blends of up to 15% bioethanol with petroleum/gasoline.

Advantages:

• Unlike petroleum, ethanol is a renewable resource

• Ethanol burns more cleanly in air than petroleum, producing less carbon (soot) and carbon monoxide

• The use of ethanol as opposed to petroleum could reduce carbon dioxide emissions, provided that a renewable energy resource was used to produce crops required to obtain ethanol and to distil fermented ethanol

Disadvantages:

• Ethanol has a lower heat of combustion (per mole, per unit of volume, and per unit of mass) that petroleum

• Large amounts of arable land are required to produce the crops required to obtain ethanol, leading to problems such as soil erosion, deforestation, fertilizer run-off and salinity.

• Major environmental problems would arise out of the disposal of waste fermentation liquors.

• Typical current engines would require modification to use high concentrations of ethanol

Biodiesel

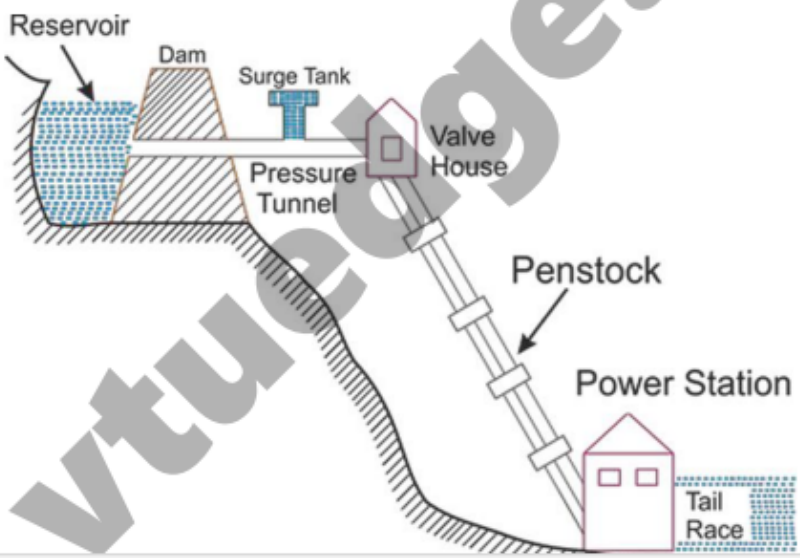
It is produced from oils or fats using transesterification and is a liquid similar in composition to fossil/mineral diesel. Chemically, it consists mostly of fatty acid methyl (or ethyl) esters (FAMEs). Feedstocks for biodiesel include animal fats, vegetable oils, soyabean, jatropha, mahua, mustard, flax, sunflower, palm oil, hemp, field pennycress, Pongamia pinnata and algae. Pure biodiesel (B100) currently reduces emissions with up to 60% compared to diesel.

Advantages of biodiesel fuel:

• An excessive production of soybeans in the world makes it an economic way to utilize this surplus for manufacturing the Biodiesel fuel.

• One of the main biodiesel fuel advantages is that it is less polluting than petroleum diesel. • The lack of sulfur in 100% biodiesel extends the life of catalytic converters.

	• Another of the advantages of biodiesel fuel is that it can also be blended with other energy resources and oil. It can also be distributed through existing diesel fuel pumps, which is another biodiesel fuel advantage over other alternative fuels. • The lubricating property of the biodiesel may lengthen the lifetime of engines. **Disadvantages of biodiesel fuel** i) At present, Biodiesel fuel is bout one and a half times more expensive than petroleum Diesel fuel. ii) It requires energy to produce biodiesel fuel from crops iii) As Biodiesel cleans the dirt from the engine, this dirt can then get collected in the fuel filter, thus clogging it. So, filters must be changed after the first several hours of biodiesel use. iv) Biodiesel fuel distribution infrastructure needs improvement. **Application of Biofuels** i) As an alternative fuel for automobiles ii) ii) Cooking purpose iii) Lubrication in place of conventional oils iv) In Power plants to generate electricity v) In marine industry to run the ships
5	Explain the different types of Fossil fuels with applications. **Fossil Fuels** : Fossil fuels are energy rich substances that have been formed from long buried plants and microorganisms. 1. **Solid Fuels:** Solid fuels are mainly classified into two categories: I. Natural fuels: wood, coal, etc. II. Manufactured fuels: charcoal, coke, briquettes, etc. **Wood**-Obtained from plants **Coal**- Charcoal is an artificial fuel obtained from wood, There are two types of coals: i) Bituminous coal ii) Anthracite coal **Applications:** used in power plants to produce electricity, Train to produce steam to run steam engine 2. **Liquid fuels**: Liquid fuels include Gasoline, Diesel oil, Kerosene, Heavy oil, Naptha, Lubricating oils, etc. These are obtained mostly by fractional distillation of crude petroleum or liquefaction of coal. **Diesel Fuel**: The diesel fuel or gas oil is obtained between 250-320oC during the fractional distillation of crude petroleum. **Petrol**: The straight run gasoline is obtained either from distillation of crude petroleum or by synthesis

	Applications: Used to run I C Engines, Power plants, Generators 3. **Gaseous fuels:** Gaseous fuels occur in nature, besides being manufactured from solid and liquid fuels. **Natural Gas**: Natural gas is generally associated with petroleum deposits and is obtained from wells dug in the oil-bearing regions. **Producer Gas**: Producer gas is essentially a mixture of combustible gases carbon monoxide and hydrogen associated with non-combustible gases N2, CO2, etc. Applications: Power plants, food processing, Garmets forcolouring the clothes etc
6	With a neat sketch Explain the working of hydel power plant. **Construction and working of Hydel power plant** A power plant that utilizes the potential energy of water for the generation of electrical energy is known as a hydroelectric power plant. Hydroelectric power plants are generally located in hilly areas where dams can be builteasily, and large water reservoirs can be made. In a hydropower plant, a water head is created by building a dam across a river or lake. From the dam, water is fed to a water turbine. **Working Principle of Hydroelectric Power Plant** The water turbine changes the kinetic energy of the falling water into mechanical energy at the turbine shaft. In simple words, falling water spins the water turbine. The turbine drives the alternator coupled with it and converts mechanical energy into electrical energy. This is the basic "working principle of hydroelectric power plant."  vtuedge.com

7	Describe the following i) Nuclear fuels ii) Fossil fuels

NUCLEAR FUELS

- ❖ Alternative source of energy.
- ❖ Uranium is the main element required to run a nuclear reactor.
- ❖ Nuclear fission or fusion will produce tremendous amount of heat energy.

Nuclear fusion: Fusion energy is a form of nuclear energy released by the fusion (combustion) of two light nuclei (i.e. nuclei of low mass) to produce heavier mass.

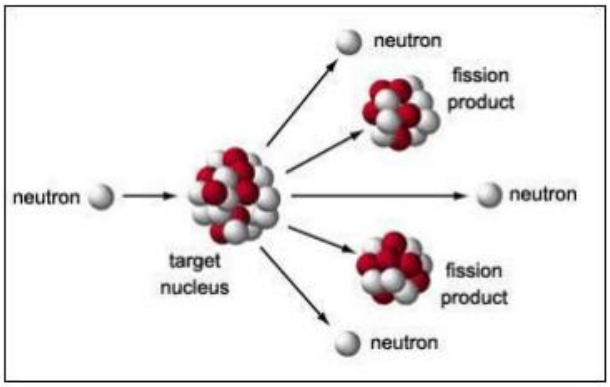
$$4_1H^1 \rightarrow 2He^4 + 2_{+1}e^0$$

Nuclear fission:

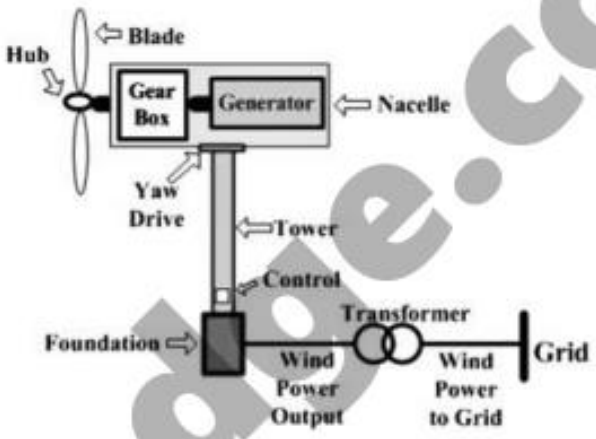
$$_{92}U^{235} + {}_0N^1 \rightarrow {}_{56}Ba^{137} + {}_{56}Kr^{97} + 2_0N^1 + \text{Energy}$$

- Nuclear fission is the process, where a heavy nucleus splits into two fragments of more or less of equal mass.

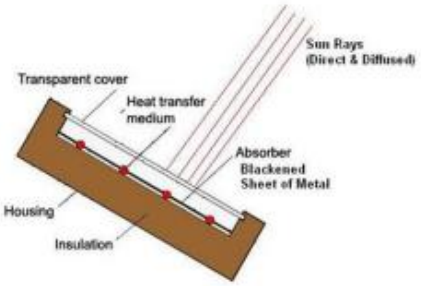
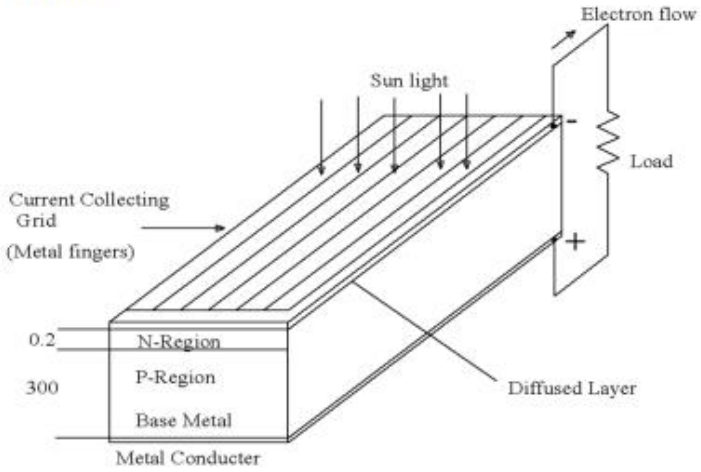
- Neutron + Heavy nucleus → Fission fragments + Neutrons (2 to 3) + energy
- The energy released by fission of I gram of U-235 is equal to that due to combustion of 50 million tons of coal ; it is about 8.5×10^{10} J.



NUCLEAR REACTOR • A nuclear reactor is a device which controls the nuclear fission chain reaction to hardness nuclear energy for peaceful purposes. • A nuclear reactor which is used t generate electricity, is called a nuclear power plant. • Fuel in the form of pellets is enclosed in several tubular claddings of steel or aluminum. This is called fuel assembly. Enriched U-235 or Pu-239 is the fuel material.

8	Describe the construction & working of wind power plant with a schematic diagram **Wind Power plant:** Wind energy is defined as the kinetic energy associated with the moment of large mass of air over the earth surface. The circulation of air is caused by the uneven heating and cooling of the earth surface. In general, during the daytime the land masses heated up faster than sea hence air moving over the land masses heated up and becomes less dense. This makes the cold denser air from the sea blow towards the land masses. But during the night, the land cools faster than the sea and hence, the denser air from the landmasses starts moving towards the sea. That is offshore.  As the wind flows over the windmill, the blades of the mill start rotating slowly. Blade starts rotated because of their aerodynamic profile. The wings are connected to hub, which is in turn connected to main driving shaft, which is intern connected to gearbox. The function of the gearbox is to increase the speed of the shaft which is connected to the generator. The generator, generates the power and transmitted to grid. The minimum velocity in air required to run the windmill is more than five meter per second.
9	Explain the utilization of solar energy using flat plate collector with a schematic diagram The collectors gather the sun‘s energy and direct it onto receivers that contain the working fluid. Basically two types of collectors are used and they are flat plate collectors and concentrating collectors. In flat plate collectors the incident solar radiation is absorbed by the collectors surface itself, which are usually coated with black paint (usually electroplated), covered with transparent glass cover on top and insulated all around to prevent the heat loss from the collector surface. The black collector surface gets heated up and then in turn transfers the heat to the fluid passing through the tubes which are either welded or soldered or are

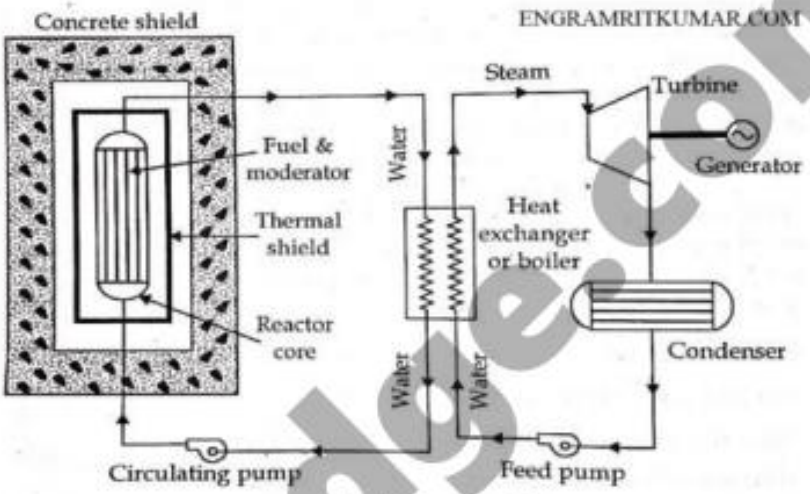
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	integral part of the collector plate. Flat plate collectors are usually sloped and oriented in one particular direction and are capable of collecting both diffuse and beam radiation. A maximum of 100°C can be easily achieved using flat plate collectors. **Flat Plate Collector** **(Non Concentrating Type)** 
10	Explain the conversion of solar energy into electricity (Photovoltaic Cell)with a schematic diagram **Helio-electrical Process:**  Conversion of solar energy into electrical energy can be achieved by the photovoltaic effect caused by the solar radiations. Photo-voltaic effect can be observed in variety of materials but best performance is given by semiconductors (like silicon). Since silicon is a tetravalent material has four valence electrons. After doping with pentavalent material (like arsenic, phosphorous) it forms N-type semiconductor and the same with trivalent material (like boron) it forms P-type semiconductor. When P-N junction of semiconductor is exposed to sunlight, a voltage around 0.5 volts is built up around the junction and current depends on the exposed area of cell. By applying external load,

	current is made to flow through the conductor, it will continue as long as the free electrons and holes are formed due to solar radiation.
11	Describe the construction & working of Nuclear power plant with a schematic diagram

Construction & Working of Nuclear Power Plant:

The schematic diagram of nuclear power station is shown in figure. A generating station in which nuclear energy is converted into electrical energy is known as nuclear power station. **The main components** of this station are nuclear reactor, heat exchanger or steam generator, steam or gas turbine, AC generator and condenser.



Nuclear power plant

- The heat liberated in the reactor due to the nuclear fission of the fuel is taken up by the coolant circulating in the reactor.
- A hot coolant leaves the reactor at top and then flows through the tubes of heat exchanger and transfers its heat to the feed water on its way.
- The steam produced in the heat exchanger is passed through the turbine and after the work has done by the expansion of steam in the turbine, steam leaves the turbine and flows to the condenser.
- The mechanical or rotating energy developed by the turbine is transferred to the generator which in turn generates the electrical energy.
- Pumps are provided to maintain the flow of coolant, condensate, and feed water.

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